PROJECT DOCUMENTATION

Solar, Wind and Biomass Energy Efficiency Analysis Using Python

A Python-based exploratory and statistical analysis of renewable energy records

Prepared by	Subramaniyam R
Dataset	Non-Conventional Energy Sources Optimizer
Dataset size	100,000 records · 11 original columns
Geographic coverage	Seven locations in India
Tools	Python · Pandas · NumPy · Matplotlib · Seaborn · Google Colab

Python | Data Cleaning | Feature Engineering | EDA | Statistical Analysis | Data Visualization

1. Project Overview

This project analyses records for three renewable energy sources—Solar, Wind and Biomass—to compare output efficiency, installation cost, annual maintenance cost and dataset-supplied optimized output. The work uses Python to inspect and prepare the dataset, engineer additional comparison features, perform exploratory and statistical analysis, and communicate findings through visualizations.

The analysis is cross-sectional. The supplied dataset does not contain a date or year column, so this project does not perform a time-trend analysis. The optimized output is treated as a supplied dataset measure, not as independently verified actual electricity generation.

2. Project Objectives

• Compare output efficiency across Solar, Wind and Biomass records.

• Examine differences in energy-related environmental conditions across locations.

• Explore how installation cost, maintenance cost and environmental variables relate to output efficiency and optimized output.

• Clean and transform the dataset using a reproducible Python workflow.

• Create meaningful features that support cost and efficiency comparisons.

• Communicate evidence-based findings and practical considerations.

3. Problem Statement

The dataset contains records for multiple renewable energy sources and locations, but the raw data alone does not make it easy to compare efficiency, costs and optimized output. This project organizes and analyses the records to identify differences and relationships across energy sources and locations. The analysis describes patterns in the supplied data; it does not establish that one variable causes another.

4. Dataset Information

Item	Details
Dataset name	Non-Conventional Energy Sources Optimizer
Original source	Hugging Face Datasets
Dataset page	huggingface.co/datasets/SivaMallikarjun/Non-Conventional-Energy-Sources-Optimizer
Domain	Renewable energy
Geographic coverage	Indian locations represented in the dataset
Time coverage	No date/year column supplied
File format	CSV
Original dimensions	100,000 rows × 11 columns
Final dimensions	100,000 rows × 15 columns

5. Dataset Attributes

Column	Description
source_type	Energy source category: Solar, Wind or Biomass
location	Location/state represented in the record
avg_temperature_c	Average temperature in degrees Celsius
wind_speed_mps	Wind speed in metres per second
solar_radiation_kwh_m2	Solar radiation in kWh per square metre
humidity_percent	Humidity percentage
altitude_m	Altitude in metres
installation_cost_usd	Installation cost in USD
maintenance_cost_usd_year	Annual maintenance cost in USD
output_efficiency_percent	Output efficiency percentage
optimized_output_kw	Dataset-supplied optimized output in kW

6. Tools and Technologies

• Python – programming and analysis.

• Pandas – data loading, cleaning, grouping and summaries.

• NumPy – numerical operations.

• Matplotlib – chart creation.

• Seaborn – statistical visualizations.

• Google Colab – notebook development and execution.

7. Project Workflow

Data Loading → Initial EDA → Data Inspection → Data Cleaning → Data Validation → Feature Engineering → Final EDA → Statistical Analysis → Univariate Analysis → Bivariate Analysis → Multivariate Analysis → Business Insights

8. Data Preprocessing and Feature Engineering

Initial inspection: The original dataset contains 100,000 rows and 11 columns. The notebook checked the dataset shape, column names, data types, descriptive statistics, missing values, duplicates and unique category values.

Data cleaning results

Check	Notebook result
Duplicate rows	0 found; no rows removed
Missing values	0 found; no imputation required
Column names	Standardized to lowercase naming
Categorical fields	source_type and location converted to category
Numeric conversion	No missing values created during conversion
IQR outlier check	0 flagged across the checked numeric columns; no rows removed

Feature engineering

New feature	Definition / purpose
total_first_year_cost_usd	installation_cost_usd + maintenance_cost_usd_year
optimized_output_per_1000_usd	optimized_output_kw ÷ installation_cost_usd × 1,000
efficiency_band	Project-defined categories: Low (≤60%), Medium (>60% to 75%), High (>75%)
location_source	Combined location and source type for grouped analysis

Final validation

After feature engineering, the final dataset contains 100,000 rows and 15 columns, with 0 missing values and 0 duplicate rows. The efficiency-band thresholds are project-defined categories and are not presented as official industry standards.

9. Statistical Analysis

Central tendency of output efficiency

Energy source	Mean (%)	Median (%)	Mode (%)
Biomass	67.47	67.43	89.00
Solar	67.53	67.71	45.78
Wind	67.60	67.54	87.49

Interpretation: Mean efficiency values are close: Wind is 67.60%, Solar is 67.53% and Biomass is 67.47%. The small differences indicate similar average efficiency in this dataset.

Variance and standard deviation

Energy source	Variance	Standard deviation
Biomass	252.13	15.88
Solar	253.45	15.92
Wind	251.06	15.84

Interpretation: Standard deviations range from 15.84 to 15.92, showing very similar spread in efficiency values across the three energy sources.

Skewness and kurtosis

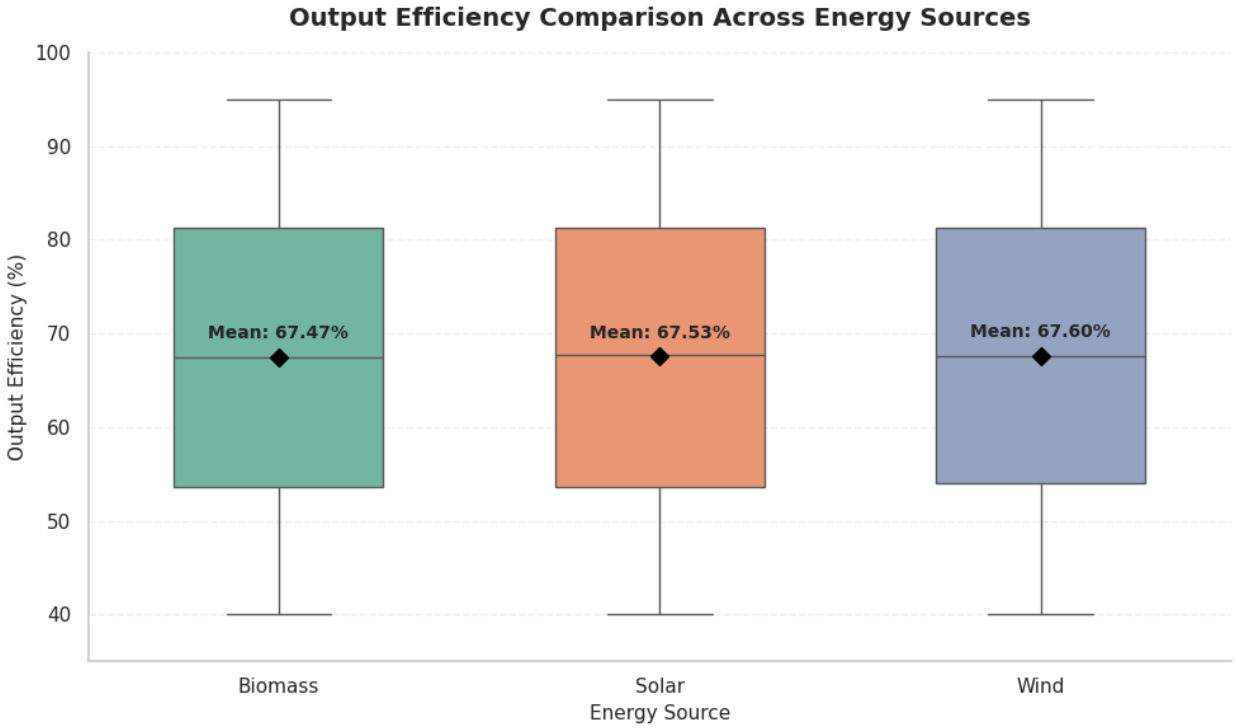
Energy source	Skewness	Kurtosis
Biomass	0.007	-1.201
Solar	-0.010	-1.206
Wind	-0.007	-1.190

Interpretation: Skewness values are close to zero and kurtosis values are close to -1.2 for all three sources, indicating broadly similar distribution shapes in this dataset.

Note: These statistics describe the supplied dataset and should not be interpreted as proof of real-world technology performance.

10. Objective-Based Visualizations and Interpretations

Chart 1 · Output Efficiency Comparison Across Energy Sources



Objective: To compare the distribution and mean output efficiency of Solar, Wind and Biomass.

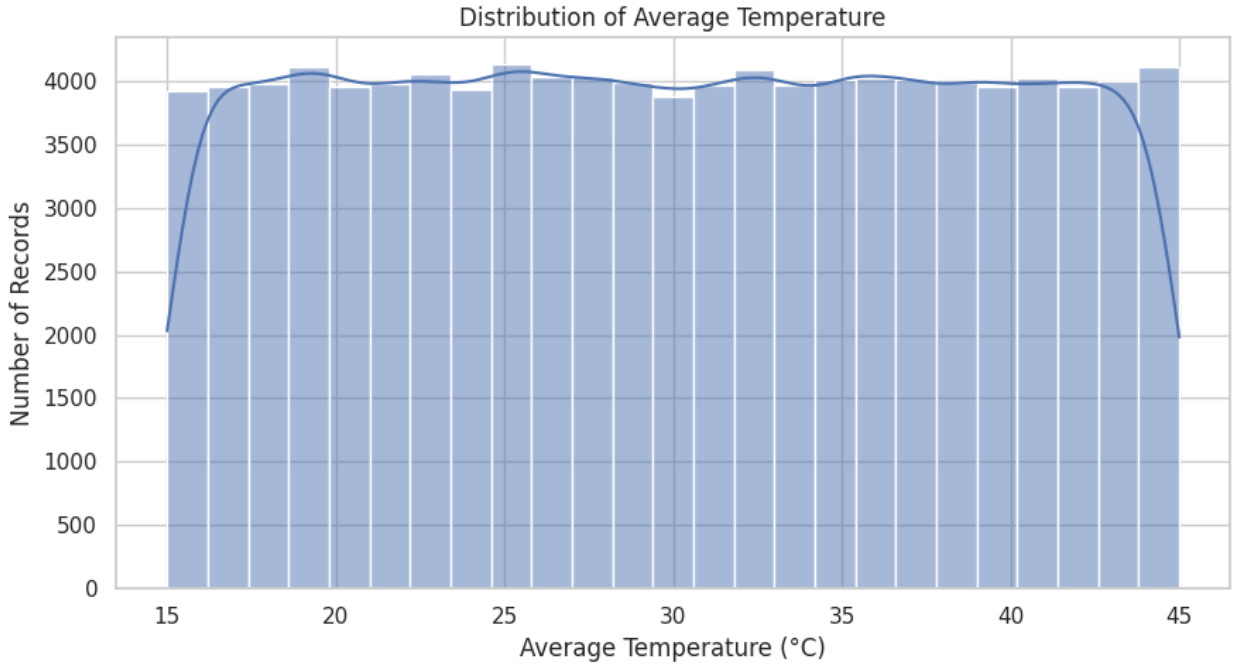
Interpretation: The boxplot shows similar efficiency distributions. Mean efficiency is 67.47% for Biomass, 67.53% for Solar and 67.60% for Wind.

Key Insight: Wind has the highest mean efficiency, but the difference between the three sources is very small.

Business Insight: Use efficiency together with cost, location and resource conditions when screening energy options; do not select a source based on this small difference alone.

Univariate Analysis

Chart 2 — Distribution of Average Temperature



Objective: To understand how average temperature values are distributed across the records.

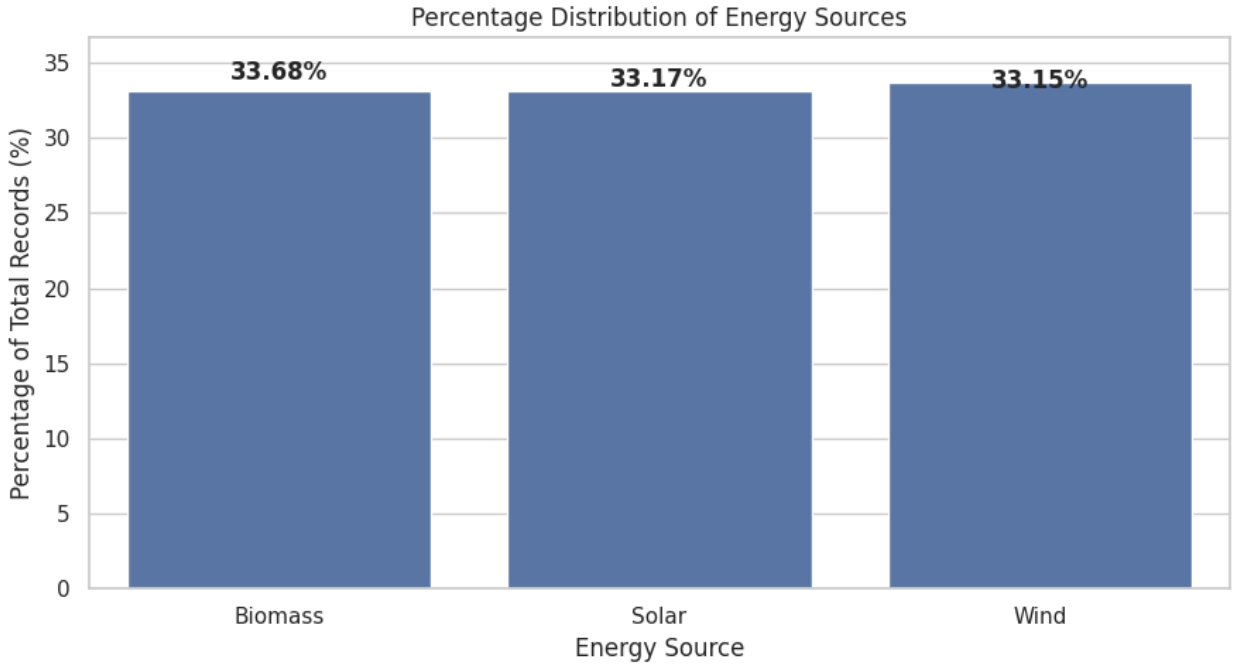
Interpretation: The histogram covers approximately 15°C to 45°C. Record counts are broadly even across the displayed temperature range, with no major gaps.

Key Insight: The dataset contains observations across a wide and fairly evenly represented temperature range.

Business Insight: Use local temperature conditions as one input to a site assessment, alongside technology-specific resource and engineering information.

Univariate Analysis

Chart 3 — Percentage Distribution of Records by Energy Source



Objective: To compare how the 100,000 records are represented across the three energy sources.

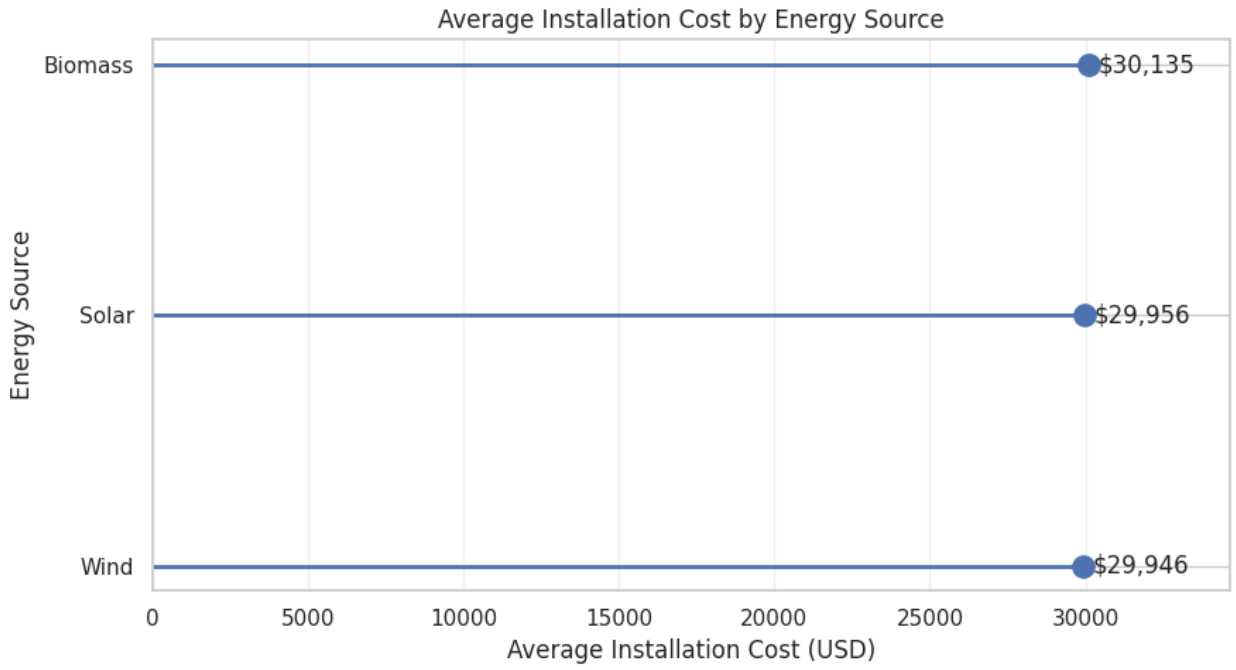
Interpretation: Biomass accounts for 33.68% of records, Solar 33.17% and Wind 33.15%. The distribution is nearly balanced.

Key Insight: No single energy source dominates the dataset by record count.

Business Insight: The balanced representation supports side-by-side dataset comparisons, but record share does not measure market share or actual energy production.

Univariate Analysis

Chart 4 — Average Installation Cost by Energy Source



Objective: To compare the average installation cost recorded for each energy source.

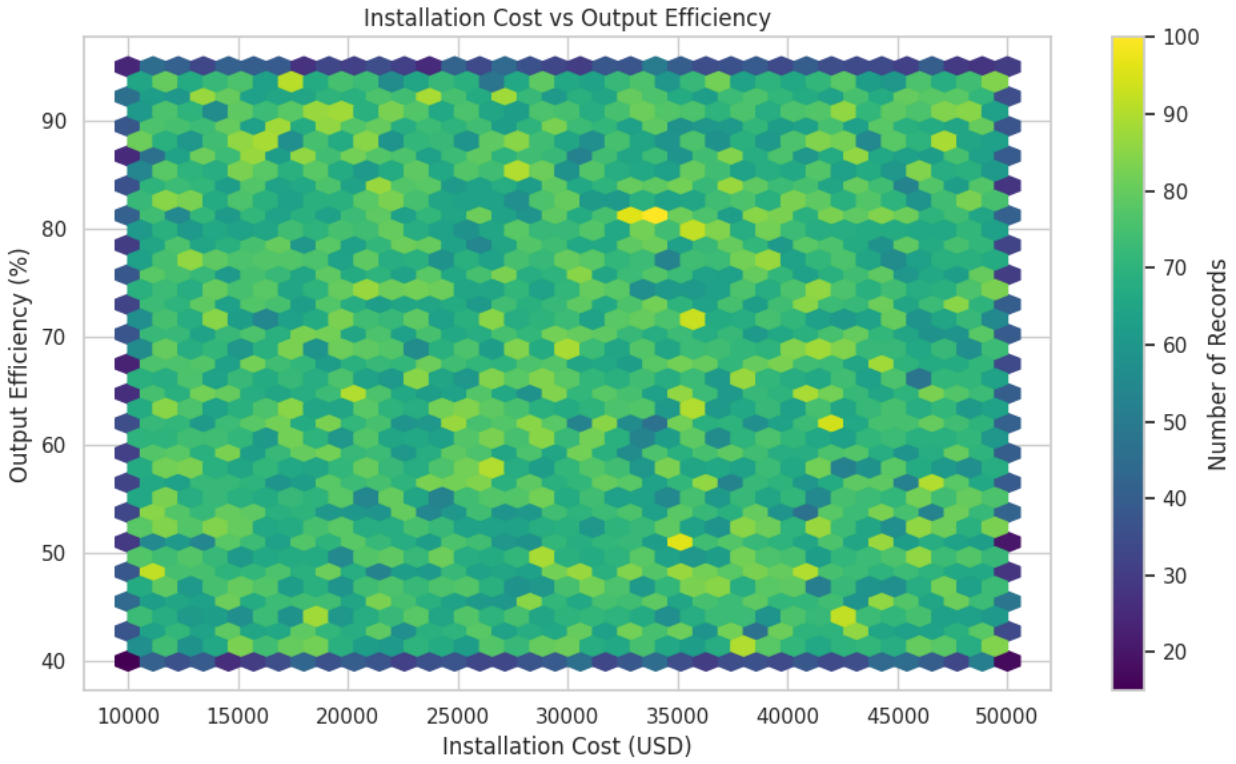
Interpretation: Biomass has the highest average installation cost at about USD 30,135. Solar is about USD 29,956 and Wind about USD 29,946.

Key Insight: Average installation costs are close, with Biomass slightly higher in this dataset.

Business Insight: Compare installation cost with annual maintenance, project lifetime and expected energy yield before making a budget decision.

Bivariate Analysis

Chart 5 — Installation Cost vs Output Efficiency



Objective: To examine the relationship between installation cost and output efficiency.

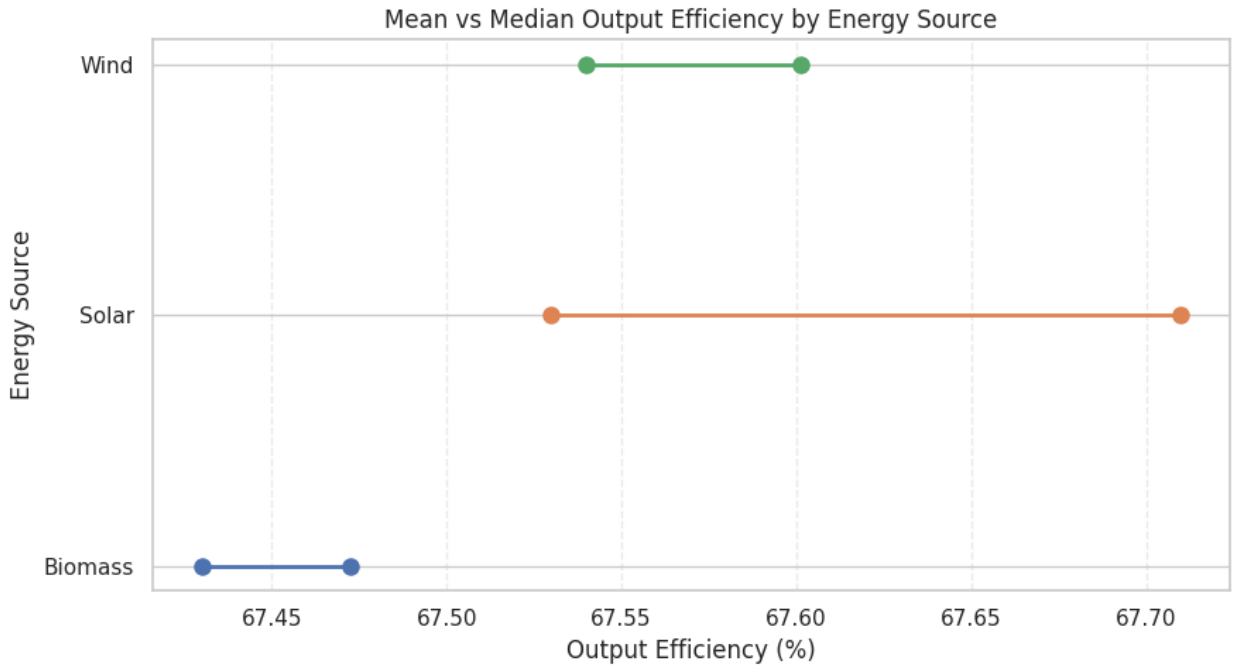
Interpretation: The hexbin plot shows records spread across cost and efficiency ranges without a clear linear pattern. Pearson correlation is -0.004.

Key Insight: Installation cost has almost no linear correlation with output efficiency in this dataset.

Business Insight: Do not assume that higher installation spending leads to higher efficiency. Evaluate technical specifications and site conditions separately.

Bivariate Analysis

Chart 6 — Mean vs Median Output Efficiency by Energy Source



Objective: To compare mean and median output efficiency for each energy source.

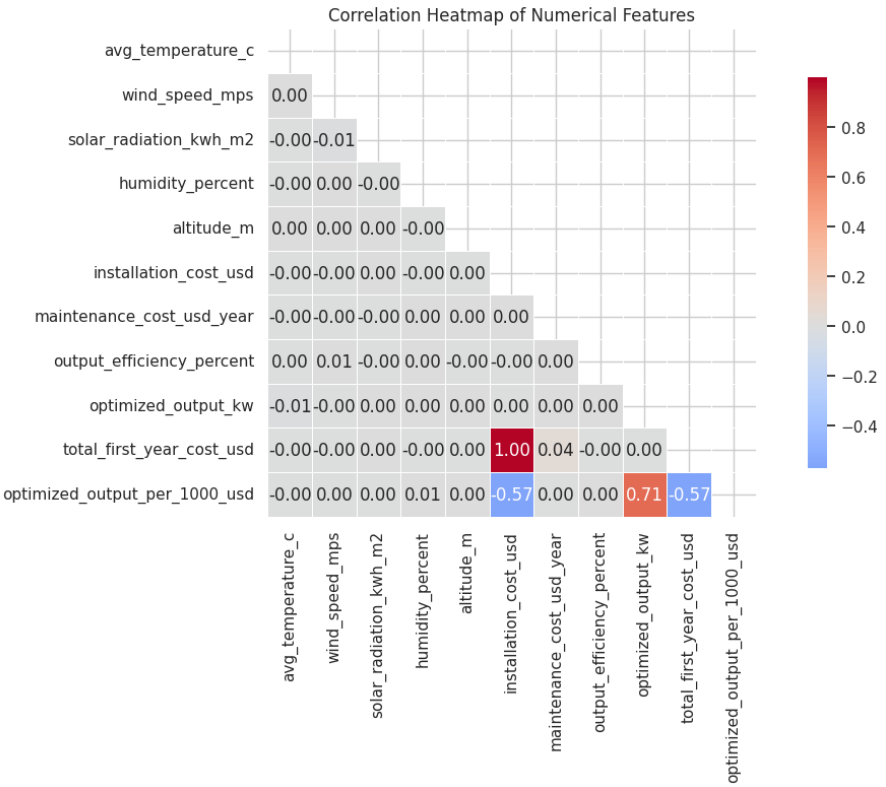
Interpretation: Biomass has mean 67.47% and median 67.43%; Solar has mean 67.53% and median 67.71%; Wind has mean 67.60% and median 67.54%.

Key Insight: Mean and median are close for all three sources, with Solar showing the largest gap among them.

Business Insight: Use both mean and median when communicating typical efficiency so that decisions are not based on a single summary measure.

Bivariate Analysis

Chart 7 — Correlation Heatmap of Numerical Features



Objective: To identify the strength and direction of linear relationships among numerical variables.

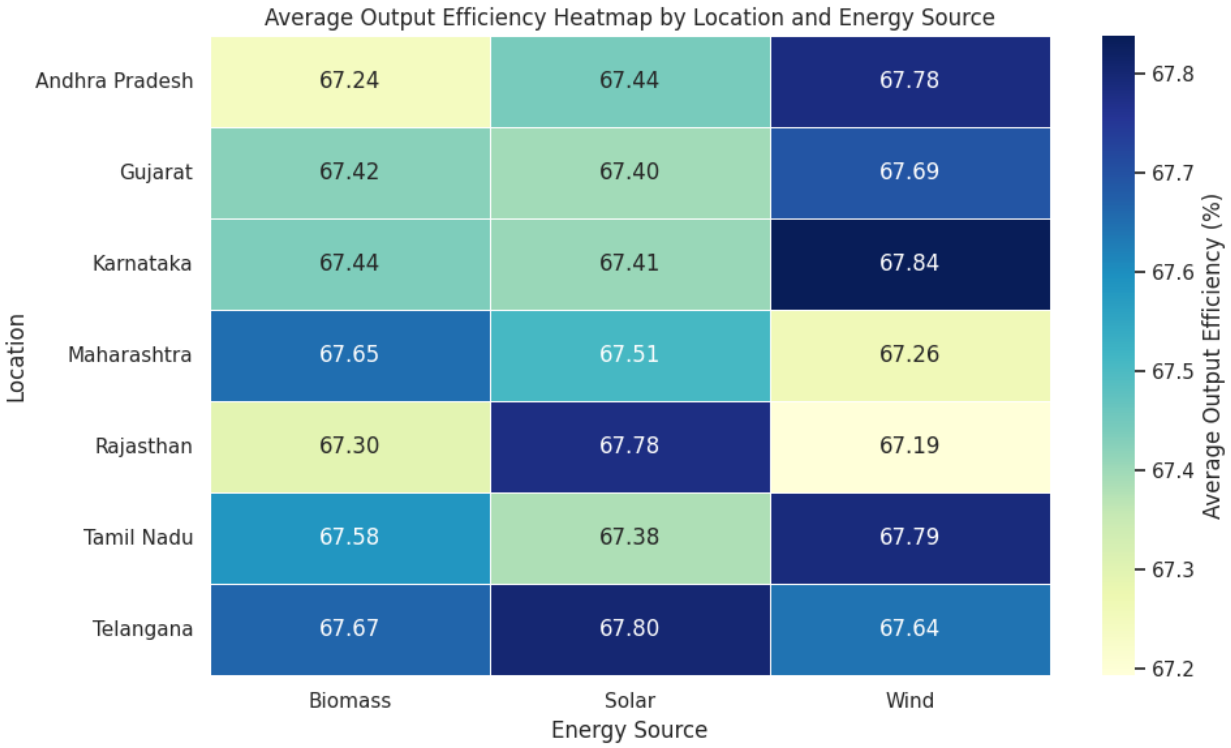
Interpretation: Optimized output and optimized output per USD 1,000 have a positive correlation of 0.71. Installation cost and optimized output per USD 1,000 have a negative correlation of -0.57.

Key Insight: The engineered output-per-cost measure is associated with optimized output and installation cost; correlation does not establish causation.

Business Insight: Use these relationships as screening signals only. Review the underlying formulas and engineering context before investment decisions.

Multivariate Analysis

Chart 8 — Average Output Efficiency by Location and Energy Source



Objective: To compare average output efficiency across seven locations and three energy sources.

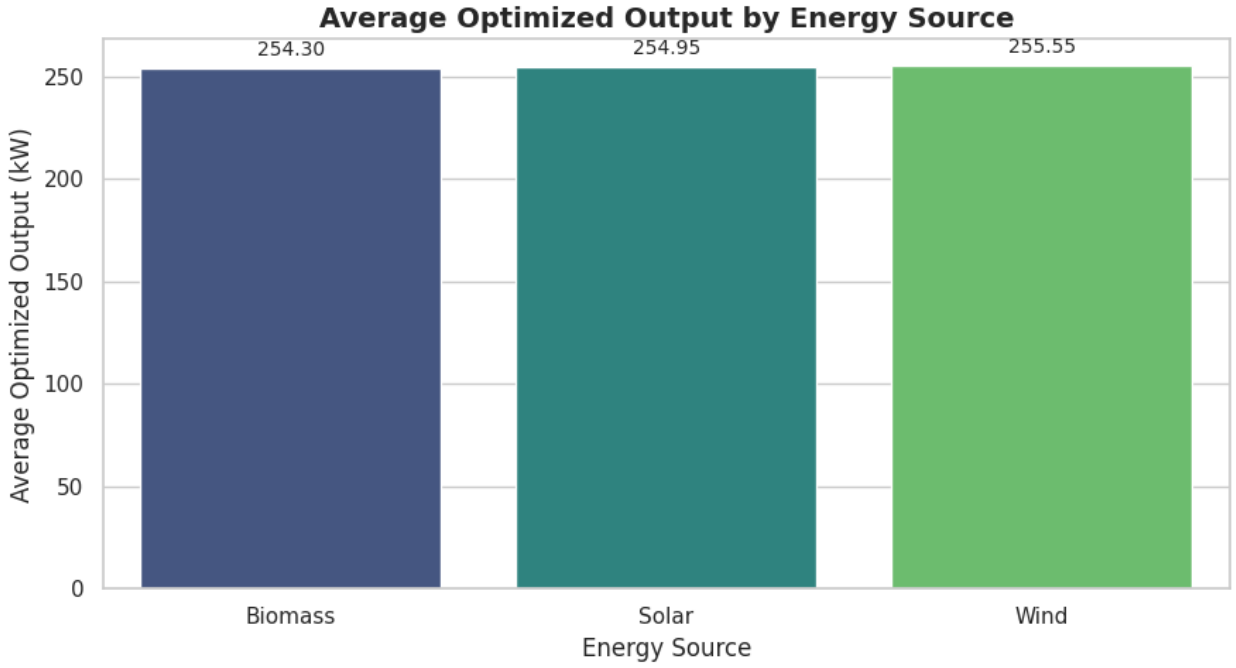
Interpretation: Wind in Karnataka records the highest displayed average at 67.84%. Biomass in Rajasthan records the lowest displayed average at 67.19%.

Key Insight: Efficiency varies by location and source, although the displayed values are relatively close.

Business Insight: Use location-level comparisons to shortlist options for detailed local resource assessment rather than treating the heatmap as a final site recommendation.

Multivariate Analysis

Chart 9 — Average Optimized Output by Energy Source



Objective: To compare the dataset-supplied average optimized output for Solar, Wind and Biomass.

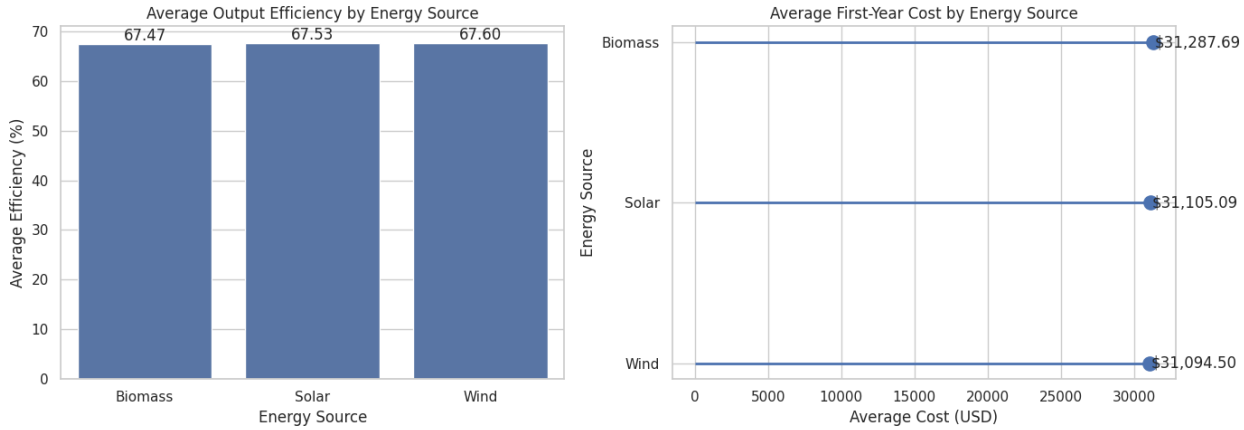
Interpretation: Wind records 255.55 kW, Solar 254.95 kW and Biomass 254.30 kW in the chart output embedded in this notebook.

Key Insight: Wind has the highest displayed average optimized output, but the differences are small.

Business Insight: Treat optimized output as a dataset-supplied measure, not verified actual generation. Validate its definition and units before using it for planning.

10. Objective-Based Visualizations and Interpretations

Chart 10 — Output Efficiency and First-Year Cost by Energy Source



Objective: To compare average output efficiency and average first-year cost together across the three sources.

Interpretation: Wind has the highest average efficiency at 67.60% and lowest average first-year cost at USD 31,094.50. Solar records 67.53% and USD 31,105.09. Biomass records 67.47% and USD 31,287.69.

Key Insight: Wind ranks slightly higher on the two displayed averages, while the differences between sources remain small.

Business Insight: Use this comparison as an initial screening view only. Consider project lifetime, financing, tariffs, reliability and verified energy-yield data before choosing a technology.

11. Summary of Findings

• Wind has the highest average output efficiency at 67.60%, followed by Solar at 67.53% and Biomass at 67.47%.

• Efficiency standard deviations are similar: Biomass 15.88, Solar 15.92 and Wind 15.84.

• Biomass has the highest average installation cost (about USD 30,135); Wind has the lowest (about USD 29,946).

• Average first-year cost is highest for Biomass at USD 31,287.69 and lowest for Wind at USD 31,094.50.

• Wind in Karnataka has the highest displayed location/source efficiency at 67.84%; Biomass in Rajasthan has the lowest at 67.19%.

• Installation cost and output efficiency have almost no linear correlation (Pearson r = -0.004).

• The dataset has no date field, so the analysis is cross-sectional rather than time-series.

12. Types of Analysis

Descriptive Analysis: Summary statistics, frequency counts and charts describe efficiency, costs, environmental variables and source representation.

Diagnostic Analysis: Grouped comparisons, hexbin plots and correlation analysis explore patterns and associations among variables. These methods do not prove causation.

Predictive Analysis: A predictive model is not included in this version of the project.

Prescriptive Analysis: Derived cost and efficiency measures support preliminary comparisons. A final investment decision requires additional verified technical and financial information.

13. Recommendations and Decision Support

• Compare energy sources using efficiency, installation cost and annual maintenance cost together.

• Use location-level results as an initial screening step, followed by local engineering and resource assessments.

• Validate the definition and units of optimized output with the dataset publisher before treating it as an energy-generation estimate.

• Avoid selecting a technology based only on small differences in average efficiency.

• For stronger economic comparisons, include project lifetime, capacity, energy yield, financing and tariff information.

14. Conclusion

Python provided a structured workflow for inspecting, cleaning and transforming the renewable-energy dataset, engineering cost and efficiency indicators, calculating statistical summaries and comparing Solar, Wind and Biomass records through visualizations. In this dataset, the three sources show very similar average efficiency. Wind is slightly higher on average efficiency and optimized output, while Biomass has the highest average installation and first-year cost. These findings describe the supplied records and should be validated with reliable engineering and financial information before real-world decisions.

15. Future Enhancements

• Add verified time-series generation data to analyze annual trends.

• Validate dataset definitions, units and methodology with the original publisher.

• Include project lifetime, installed capacity and measured energy-yield information for stronger cost comparisons.

• Build an interactive dashboard in Power BI or Tableau.

• Develop a predictive model only after defining a suitable target, checking assumptions and validating performance against a baseline.

16. Additional Comparative Findings

This section brings together the key cost and location comparisons from the notebook to make the results easier to interpret alongside the visualizations.

Average installation and annual maintenance cost

Energy source	Avg. installation cost (USD)	Avg. annual maintenance (USD)	Avg. first-year cost (USD)
Biomass	30,135	1,152.94	31,287.69
Solar	29,956	1,149.21	31,105.09
Wind	29,946	1,148.23	31,094.50

Interpretation: Biomass has the highest average installation cost and first-year cost in this dataset. Wind has the lowest average installation cost and first-year cost. Annual maintenance averages are close across all three sources.

Key insight: The average first-year cost difference between Biomass and Wind is about USD 193.19. This is a dataset comparison and does not include lifetime operating costs or financing.

Business insight: Compare upfront and annual costs together, then extend the analysis to project lifetime, capacity, tariffs, financing and verified energy yield before selecting a technology.

Location coverage

Location	Record count
Karnataka	14,432
Tamil Nadu	14,417
Andhra Pradesh	14,361
Gujarat	14,239
Maharashtra	14,184
Rajasthan	14,208
Telangana	14,159

Interpretation: The seven locations have broadly similar record counts, each contributing around 14,000 records to the 100,000-row dataset.

Key insight: No single location dominates the dataset by record count.

Business insight: This relatively balanced coverage allows location-level comparisons in the dataset. It does not guarantee that the sample represents the actual installed renewable-energy market in each state.

17. Methodology, Scope and Limitations

Analysis methodology

• Data inspection: reviewed dataset dimensions, field names, data types, summary statistics and category values.

• Data quality checks: checked missing values and duplicate records before analysis.

• Data preparation: standardized field names and validated numeric and categorical data types.

• Feature engineering: created first-year cost, optimized output per USD 1,000, efficiency band and location-source combination.

• Exploratory analysis: used univariate, bivariate and multivariate charts to study distributions, comparisons and relationships.

• Statistical analysis: calculated mean, median, mode, variance, standard deviation, skewness, kurtosis and Pearson correlation where applicable.

Scope and limitations

• The dataset contains 100,000 records and 11 original columns; the engineered dataset contains 15 columns.

• The dataset includes Solar, Wind and Biomass records across seven Indian locations.

• No date or year column is available, so yearly trend analysis is outside the scope of this project.

• The optimized_output_kw field is treated as a dataset-provided optimized measure. It has not been independently validated as actual measured electricity generation.

• Correlation describes association and does not establish cause and effect.

• The analysis is based on the supplied dataset and should not be treated as a complete technical feasibility study or investment recommendation.

• The efficiency band cutoffs used in feature engineering are project-defined for analysis and are not claimed to be official industry standards.

Final conclusion

The analysis provides a structured comparison of renewable-energy records using efficiency, cost, optimized output and location. Average efficiency values are close across the three sources. Wind is slightly higher on the dataset's average efficiency and optimized-output measures, while Biomass has the highest average installation and first-year cost. The results are useful for preliminary comparison, but real-world selection requires validated engineering, generation and financial data.

Notebook chart-title and value verification

The notebook's boxplot title is **Output Efficiency Comparison Across Energy Sources**; the report describes it as an output-efficiency distribution boxplot. The temperature histogram is titled **Distribution of Average Temperature**. The optimized-output chart labels read Biomass 254.30 kW, Solar 254.95 kW and Wind 255.55 kW. The report uses these values from the actual embedded chart image.

The project is exploratory and statistical. No predictive machine-learning model is included in this documentation.